

VISUAL ANALYTICS & INSIGHTS

using Power BI

Session 2 | Duration: 2 Hour

MsBelindaAllen — Accounting Professionals Series

About Belinda Allen

From Alcolu, SC

- I lived the longest in New York City

In Technology

- Principal Enablement Consultant – iLink Digital
- Started in tech in 1991
- Microsoft Partner since 2001

Microsoft MVP

- 14 years / 5 years GP and 9 years Power BI

Microsoft MCT

- 15 years (with a hiatus in the middle)

Co-Authored 2 books

- BI and Dynamics GP

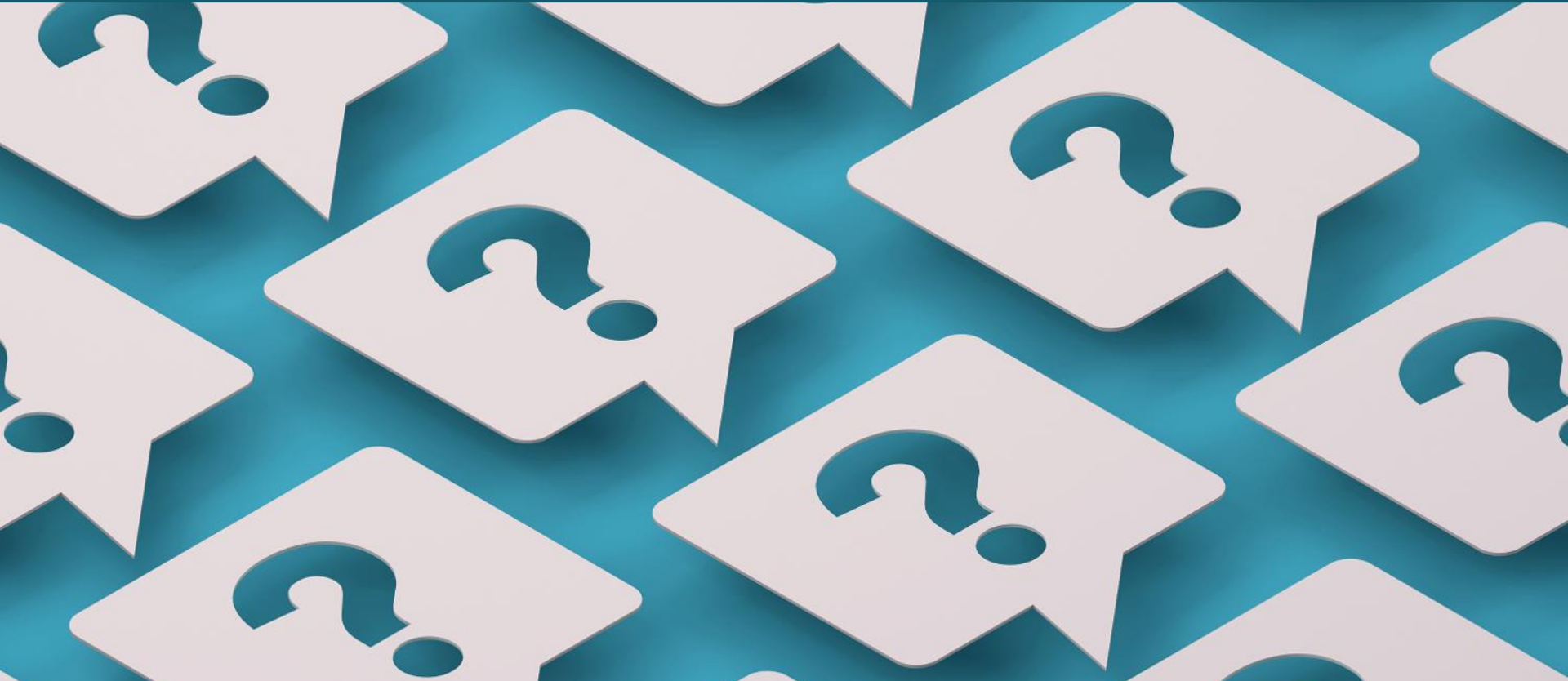
I love learning

- Technology, Wood Turning, Cooking

I love hobbies

- Knitting/Crocheting, Reiki

Poll questions



WORKSHOP AGENDA

Intro	Session 2 Welcome & Learning Objectives
1	Module 1 — Visual Best Practices
2	Module 2 — Charts, Slicers & Drill-Through
3	Module 3 — KPIs, Conditional Formatting & Storytelling
4	Module 4 — Demo: Executive Sales Dashboard
Close	Wrap-Up, Resources & Q&A

SESSION 2 LEARNING OBJECTIVES

- Identify the right chart type for each financial reporting scenario
- Build interactive slicers, drill-through pages, and bookmarks
- Apply conditional formatting to surface outliers instantly
- Design KPI cards and scorecards that drive decisions
- Tell a compelling data story with a polished executive dashboard
- Customize themes and color schemes to match client branding



MODULE 1

VISUAL BEST PRACTICES

Choosing the Right Visual for Every Financial Story



CHOOSING THE RIGHT CHART TYPE

MODULE 1

Bar / Column Chart

Comparing values across categories: revenue by dept, expenses by month.

Line Chart

Trends over time: monthly cash flow, rolling 12-month revenue.

Pie / Donut

Part-to-whole: % of revenue by client. Use sparingly — max 5 slices.

Scatter Chart

Relationship between two variables: budget vs. actual by project.

Table / Matrix

Detailed GL data with row-level drill. Replaces exported Excel reports.

Treemap

Hierarchical spending: division → department → account.

Card / KPI

Single-number callouts: Total Revenue, Net Income, DSO.

Waterfall

Cumulative changes: bridge from prior-year to current-year income.

VISUAL DESIGN PRINCIPLES FOR ACCOUNTANTS

MODULE 1

1. One Message Per Visual

Each chart should answer a single question. If a visual needs a paragraph of explanation, redesign it.

2. Color with Purpose

Use color to highlight, not decorate. Green = favorable, Red = unfavorable. Limit to 3–4 colors per report.

3. Context is Everything

Always show a benchmark: prior year, budget, or target. A number without context is just noise.

4. Less is More

Remove gridlines, borders, and data labels that don't add meaning. White space is not wasted space.

MODULE 2

CHARTS, SLICERS & DRILL-THROUGH

Interactive Features That Make Reports Come Alive



SLICERS — YOUR REPORT'S CONTROL PANEL

MODULE 2

1

Add a Slicer

Visualizations pane → Slicer icon. Drop in a field: Date, Department, Region, or Account Type.

2

Format It

Slicer header, font, background. Switch between List, Dropdown, or Between (for date ranges).

3

Sync Across Pages

View → Sync Slicers. One date picker controls every page — no manual replication.

4

Cross-Filtering

Click any bar or slice and every other visual filters automatically. No extra configuration needed.



Pro tip: Use a Dropdown slicer on mobile layouts to save space. Use Between for fiscal-year date ranges.

Drill-Down (Within a Visual)

Click the expand arrows on a hierarchy visual. Year → Quarter → Month → Day. Works on bar charts, matrices, and treemaps.

Drill-Through (Cross-Page Navigation)

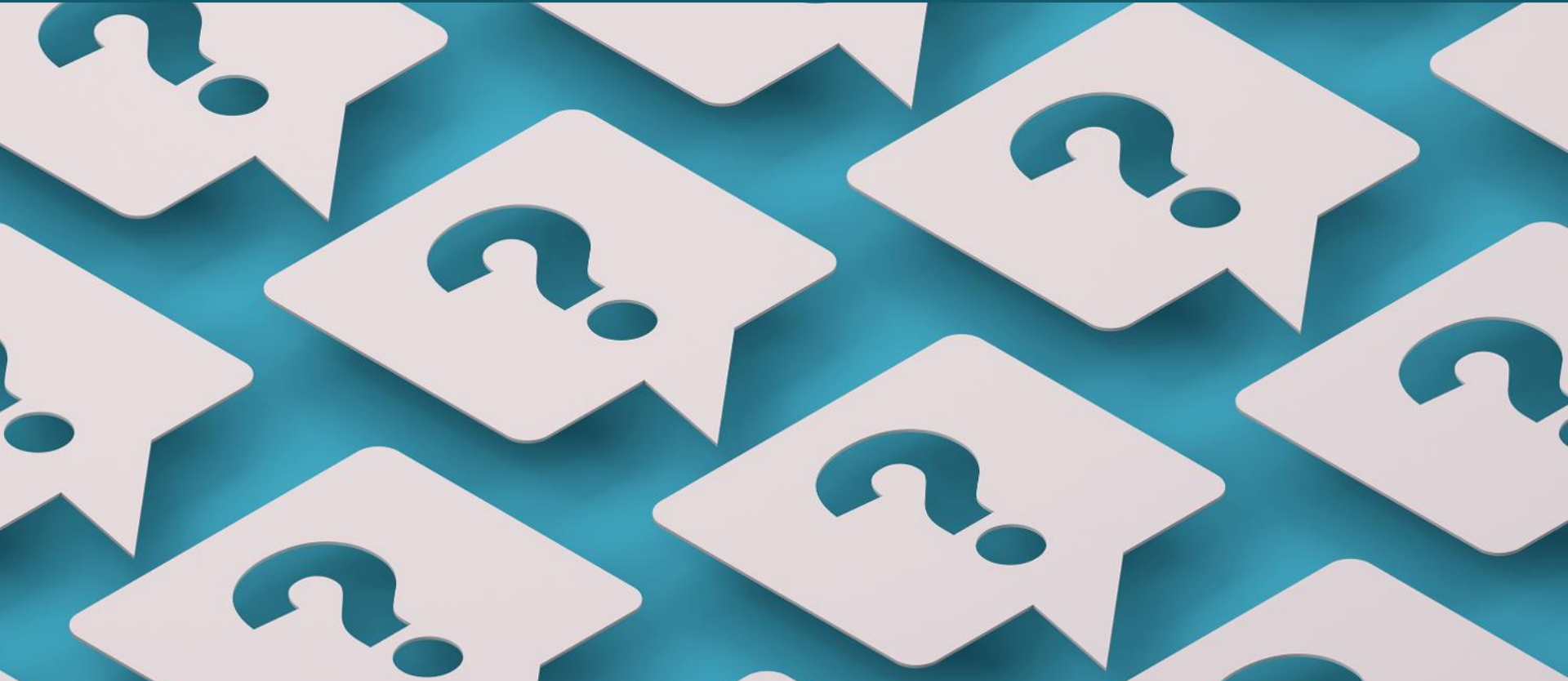
Right-click any data point → Drill through → select a detail page. Automatically filters the destination page by the selected value. Perfect for GL transaction detail.

💡 **Bookmarks:** Capture a specific filter state (e.g. “Q4 View” or “Audit Mode”) and assign to a button. One click restores the exact view — great for client presentations.

HOW TO SET UP DRILL-THROUGH

1. Create a detail page (e.g. “GL Detail”)
2. Drag target field into “Drill-through filters” well
3. Back button appears automatically
4. Right-click any bar → Drill through → GL Detail

Poll questions



MODULE 3

KPIs, CONDITIONAL FORMATTING & STORYTELLING

Making Data Impossible to Ignore



// Core Revenue Measures

```
Total Revenue = SUM(Sales[Revenue])
```

```
Total Cost = SUM(Sales[Cost])
```

```
Gross Profit = [Total Revenue] - [Total Cost]
```

```
GP % = DIVIDE([Gross Profit], [Total Revenue], 0)
```

```
Prior Year Revenue = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR('Date'[Date]))
```

```
Revenue Variance % = DIVIDE([Total Revenue] - [Prior Year Revenue], [Prior Year Revenue])
```

💡 Use `DIVIDE()` instead of `/` to handle zero-division gracefully. Always test with a blank slicer selection to confirm your measures return `BLANK()` and not `0` when there's no data.

KPI CARDS & SCORECARDS

MODULE 3

TOTAL REVENUE

\$2.4M

vs. \$2.1M prior year ↑ 14%

TOTAL EXPENSES

\$1.8M

vs. \$1.7M prior year ↑ 6%

NET INCOME

\$620K

vs. \$400K prior year ↑ 55%

EBITDA MARGIN

25.8%

Target: 22% ✓ Above target

HOW TO BUILD A KPI CARD IN POWER BI

1. Select “Card” visual → drag in your measure (e.g. [Total Revenue]) • 2. Format → Data label: set font, size, color
3. Format → Category label: rename or hide • 4. Add conditional color: Format → Data label → Conditional formatting → By value

SCORECARD TIP: Use a Matrix visual with conditional formatting icons (✓ ▲ ▼) for a full management dashboard.

DAX for variance %:

Revenue Variance % = `DIVIDE([Total Revenue] - [Prior Year Revenue], [Prior Year Revenue])`

CONDITIONAL FORMATTING



Value < Budget by >10%

Red fill + ▼ icon. Draws immediate attention to significant unfavorable variances.



Value within 5% of Budget

Yellow / amber fill. Signals caution without triggering alarm.



Value ≥ Budget or Target

Green fill + ▲ icon. Confirms favorable performance at a glance.



Data bar (background)

Horizontal bar fills cell proportionally. Best for ranking columns in tables.

WHERE TO APPLY: Right-click column header in table → Conditional formatting → Background color / Icon sets / Data bars

DAX color measure example:

```
Variance Color = IF([Budget Variance %] < -0.1, "#C00000", IF([Budget Variance %] < 0, "#FFC000", "#00B050"))
```

We'll be using Revenue Variance % rather than Budget Variance.

The “So What” Framework: Every number should answer a business question.

SITUATION

“Q3 revenue came in at \$2.4M — 14% above prior year.”

COMPLICATION

“But operating expenses grew 18%, compressing net margin from 22% to 19%.”

RESOLUTION

“The payroll spike in Ops drove \$180K of excess spend. Recommend reviewing headcount plan for Q4.”

DESIGN TIP: Use a text box on a blank page as a “Title page” or “Executive Summary” with the key takeaway in 1–2 sentences. Link to detail pages via bookmarks. Your CFO reads the summary — the auditor reads the detail.



MODULE 4

DEMO: EXECUTIVE SALES DASHBOARD

Build Your First Executive-Grade Report



DEMO OVERVIEW — EXECUTIVE SALES DASHBOARD

MODULE 4

STEP 1 / 5

Import Sales Data

Excel file with columns: Date, Region, Salesperson, Product, Units, Revenue, Cost

STEP 2 / 5

Build Date Table

CALENDARAUTO() + fiscal year columns. Relate to Sales[Date]. Enable “Mark as Date Table”.

STEP 3 / 5

Write Core Measures

Total Revenue, Total Cost, Gross Profit, GP%, Prior Year Revenue, Revenue Variance %

STEP 4 / 5

Build the Dashboard Page

KPI cards (row 1) + Line chart (revenue trend) + Bar chart (by region) + Slicer (year/region)

STEP 5 / 5

Add Drill-Through Detail

New page: Salesperson detail table with conditional formatting. Link from region bar chart.

EXECUTIVE SALES DASHBOARD — LAYOUT

MODULE 4

Executive Sales Dashboard | FY 2024

TOTAL REVENUE

\$4.8M

GROSS PROFIT

\$1.9M

GP MARGIN

39.6%

YoY VARIANCE

+12.4% ▲

Revenue by Month (Line Chart)

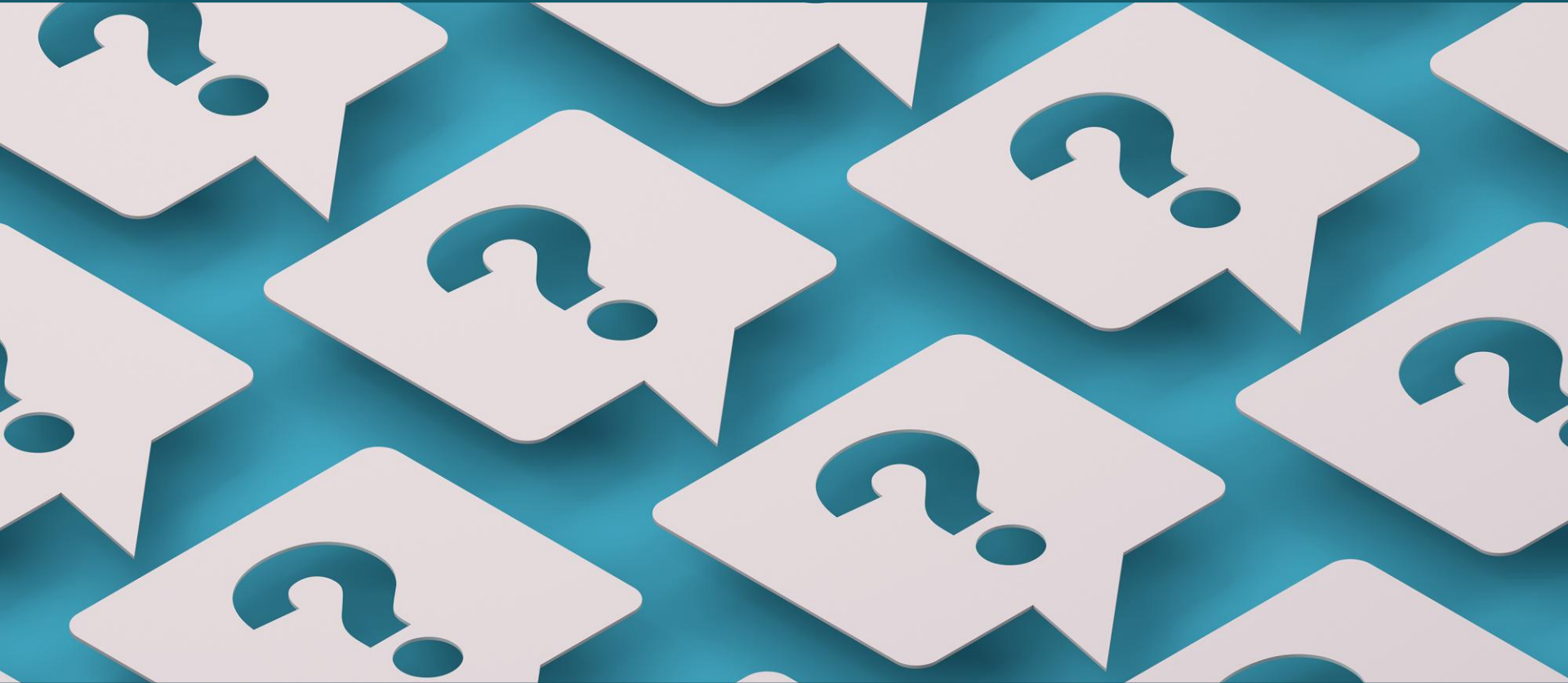
Revenue by Department (Bar Chart)

Salesperson Detail Table (with conditional formatting on Variance %) — Drill-Through from Region Chart

Slicer: Year • Department • Sub-Account



Poll questions



SESSION RECAP

Module 1

Choose chart types strategically. Bar for comparisons, Line for trends, KPI cards for callouts. Design with purpose: one message per visual.

Module 2

Slicers, drill-through, and bookmarks turn a static report into an interactive app. Sync slicers across pages for consistency.

Module 3

Conditional formatting reveals outliers instantly. KPI scorecards + the So What framework turn data into insight-driven narratives.

Module 4

Build live: import → date table → DAX measures → dashboard layout → drill-through detail. Polish with themes and color.

NEXT STEPS & RESOURCES

1	Download the sample Sales Dashboard dataset at powerbi.microsoft.com/sample-datasets
2	Complete “Create and Use Analytics Reports with Power BI” on Microsoft Learn (free, 5 hrs)
3	Practice: Recreate today’s dashboard using your own firm’s Excel data
4	Explore Power BI Themes: powerbi.microsoft.com/themes-gallery — apply client branding in one click
5	Session 3 preview: Power BI for Audit — testing with DAX, anomaly detection, and audit trails

Thank You!

Q&A

Questions & Discussion

Discussion prompts (if needed):

1. Which chart type surprised you most today?
2. What report would you build first for your team or clients?
3. What's one Power BI visual you want to try this week?